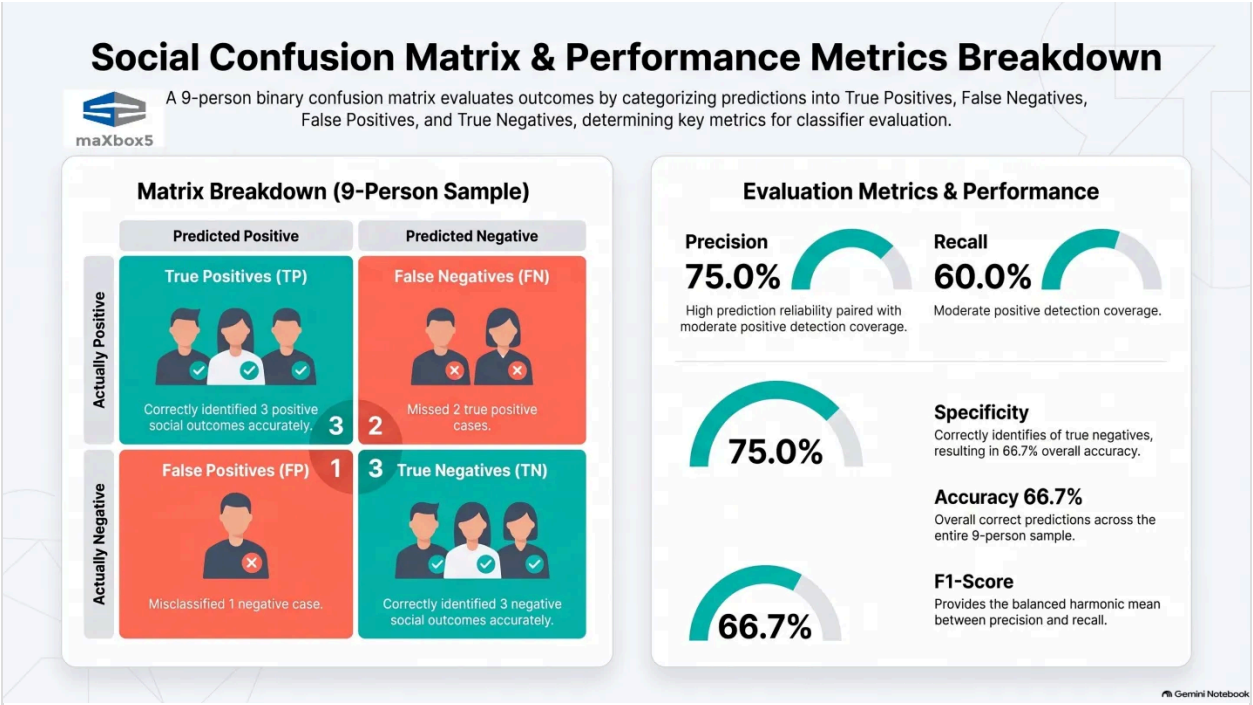


Behind the Confusion Matrix

October 19, 2026

Study Guide: Person A's 9-Person Social Confusion Matrix & Performance Metrics, Tutorial 203

1. Introduction to the Social Confusion Matrix

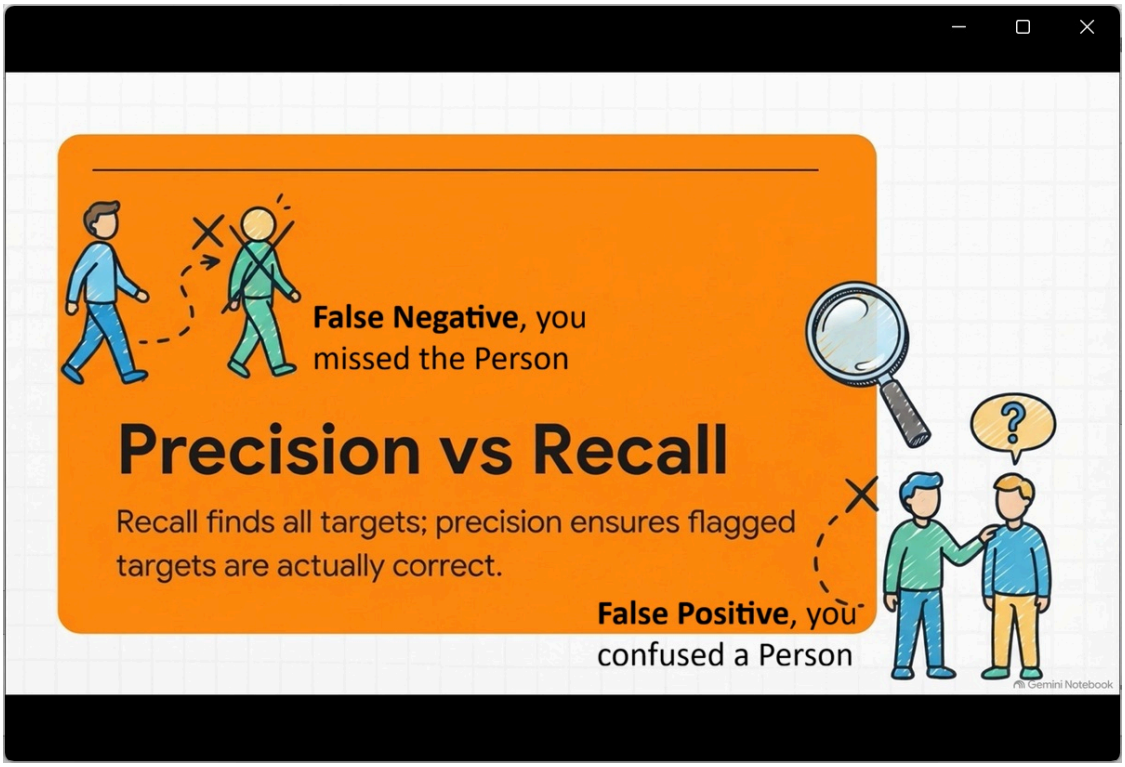
A confusion matrix is an empirical evaluation tool used to categorize binary classification outcomes by comparing ground-truth reality against predicted outcomes. In social recognition and outcome modeling, classification performance reflects how accurately an individual or system identifies social entities or states.

The fundamental intuitive principles governing social recognition metrics are:

- Recall Deficit (Bad Recall or False Negative): Occurs when a person or positive social case is present but fails to be recognized or detected (a missed person).
- Precision Deficit (Bad Precision or False Positive): Occurs when a person or case is confused with another, or when a negative case is incorrectly identified as positive (a false alarm or confused).

This study guide analyzes Person A's 9-person social recognition dataset, evaluating its 2 \times 2 structure, mathematical derivations, real-world interpretations, and performance trade-offs. In the 9-person social recognition

scenario, Person A evaluates 9 total individuals across two ground-truth categories: Actually Positive (people Person A actually knows) and Actually Negative (strangers):



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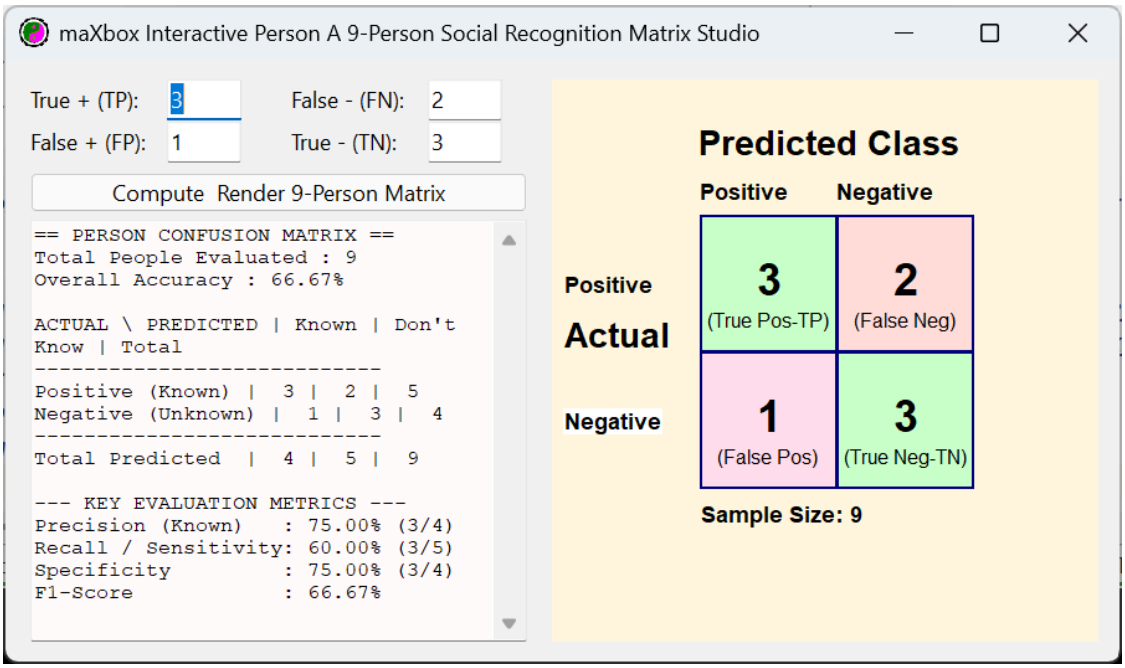
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2. Structural Breakdown of Person A's 2×2 Matrix

Person A's social recognition scenario evaluates a sample of 9 total individuals. The system's predictions are categorized across two dimensions: Actual Class (Ground Truth) and Predicted Class (Decision/Outcome).



https://sourceforge.net/projects/maxbox5/files/examples/1503_person_the_confusion_matrix2_1interactiv.pas/download

Matrix Distribution Table

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Actual \ Predicted	Predicted: Known	Predicted: Don't Know	Total Actual
Actually Known	3 (TP)	2 (FN)	5
Actually Don't Know	1 (FP)	3 (TN)	4
Total Predicted	4	5	9

	Predicted Positive	Predicted Negative	Row Total (Actual)

Actually Positive	True Positives (TP): 3	False Negatives (FN): 2	5
Actually Negative	False Positives (FP): 1	True Negatives (TN): 3	4
Column Total (Predicted)	4	5	Total Sample: 9

Breakdown of Matrix Quadrants

1. True Positives (TP = 3):
    - *Definition:* Correctly identified positive social outcomes.
    - *Person A Scenario:* Person A accurately identified 3 positive cases out of the sample.
2. False Negatives (FN = 2):
    - *Definition:* Actual positive cases that were incorrectly predicted as negative (misses).
    - *Person A Scenario:* Person A missed 2 true positive cases, failing to recognize them when they were actually present.
3. False Positives (FP = 1):
    - *Definition:* Actual negative cases that were misclassified as positive (false alarms/confusion).
    - *Person A Scenario:* Person A misclassified 1 negative case as positive.
4. True Negatives (TN = 3):
    - *Definition:* Correctly identified negative social outcomes.
    - *Person A Scenario:* Person A accurately identified 3 negative cases.

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3. Mathematical Formulas, Calculations, and Metric Interpretations

Using the primary parameters (TP = 3, FN = 2, FP = 1, TN = 3), the key performance evaluation metrics are derived below.

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{3}{3 + 1} = \frac{3}{4} = 75.0\%$$

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{3}{3 + 2} = \frac{3}{5} = 60.0\%$$

$$\text{Specificity} = \frac{TN}{TN + FP} = \frac{3}{3 + 1} = \frac{3}{4} = 75.0\%$$

$$\text{Accuracy} = \frac{TP + TN}{\text{Total Sample}} = \frac{3 + 3}{9} = \frac{6}{9} = 66.7\%$$

$$\text{F1-Score} = \frac{2 * \text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2 * 0.750 * 0.600}{0.750 + 0.600} = \frac{0.900}{1.350} = 66.7\%$$

Detailed Evaluation of Metrics

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3. Key Performance Metrics

- **Accuracy:** $\frac{TP+TN}{Total} = \frac{3+3}{9} = \frac{6}{9} \approx \mathbf{66.7\%}$
- **Precision:** $\frac{TP}{TP+FP} = \frac{3}{3+1} = \frac{3}{4} = \mathbf{75.0\%}$ (When A says they know someone, they are right 75% of the time)
- **Recall:** $\frac{TP}{TP+FN} = \frac{3}{3+2} = \frac{3}{5} = \mathbf{60.0\%}$ (A correctly recognizes 60% of all people they actually know)
- **F1-Score:**
  $2 \cdot \frac{Precision \cdot Recall}{Precision+Recall} = 2 \cdot \frac{0.75 \cdot 0.60}{0.75+0.60} \approx \mathbf{66.7\%}$

1. Precision (Positive Predictive Value – PPV)

- Formula: \text{Precision} = \frac{TP}{TP + FP}
- Calculation: \frac{3}{3 + 1} = \frac{3}{4} = 0.750 \text{ or } 75.0\%
- Interpretation: Precision evaluates prediction reliability. Out of all individuals Person A flagged as positive, 75.0% were genuinely positive. This demonstrates high prediction reliability when an assertion is made.

2. Recall (True Positive Rate / Sensitivity)

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- Formula: \text{Recall} = \frac{TP}{TP + FN}
- Calculation: \frac{3}{3 + 2} = \frac{3}{5} = 0.600 \text{ or } 60.0\%
- Interpretation: Recall measures detection coverage. Person A successfully detected **60.0%** of the actual positive cases in the sample, missing 40.0% (FN = 2). This indicates moderate positive detection coverage.

3. Specificity (True Negative Rate – TNR)

- Formula: \text{Specificity} = \frac{TN}{TN + FP}
- Calculation: \frac{3}{3 + 1} = \frac{3}{4} = 0.750 \text{ or } 75.0\%
- Interpretation: Specificity reflects the ability to correctly identify negative outcomes. Person A correctly identified **75.0%** of all truly negative instances.

4. False Positive Rate (FPR)

- Formula: \text{FPR} = \frac{FP}{FP + TN} = 1 – \text{Specificity}
- Calculation: \frac{1}{1 + 3} = \frac{1}{4} = 0.250 \text{ or } 25.0\%
- Interpretation: Out of all actual negative cases, **25.0%** were incorrectly flagged as positive.

5. Overall Accuracy

- Formula: \text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}
- Calculation: \frac{3 + 3}{9} = \frac{6}{9} \approx 0.6667 \text{ or } 66.7\%
- Interpretation: Across the entire 9-person sample, Person A made overall correct predictions (both positive and negative) in **66.7%** of cases.

6. F1-Score

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- Formula: \text{F1-Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}
- Calculation: 2 \cdot \frac{0.750 \cdot 0.600}{0.750 + 0.600} = 2 \cdot \frac{0.450}{1.350} = \frac{0.900}{1.350} \approx 0.6667 \text{ or } 66.7\%
- Interpretation: The F1-Score represents the balanced harmonic mean between precision and recall, balancing prediction reliability against detection coverage to yield a single score of 66.7%.

4. Summary of Metrics & Performance Profile

The evaluation metrics for Person A's 9-person social recognition sample are summarized below:

Metric	Exact Fractional Value	Percentage	Core Interpretation
Precision	3/4	75.0%	High prediction reliability when claiming a positive match.
Recall	3/5	60.0%	Moderate positive detection coverage; misses 2 out of 5 actual positive cases.
Specificity	3/4	75.0%	Strong ability to filter out and identify true negative cases.

Accuracy	6/9	66.7%	Correct overall predictions across the total 9-person dataset.
F1-Score	0.900 / 1.350	66.7%	Balanced harmonic mean reconciling precision and recall.

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5. Theoretical Bridge: Connecting Confusion Matrix Metrics to Bayes' Theorem

Precision in a empirical confusion matrix directly corresponds to the **posterior probability** in Bayes' theorem. Bayes' theorem demonstrates how error rates (Recall and FPR) interact with the **prevalence** of positive cases in a population to determine the actual precision of positive predictions:

$$\text{Precision} = \frac{\text{Recall} \cdot \text{Prevalence}}{\text{Recall} \cdot \text{Prevalence} + \text{FPR} \cdot (1 - \text{Prevalence})}$$

Application to Person A's Sample

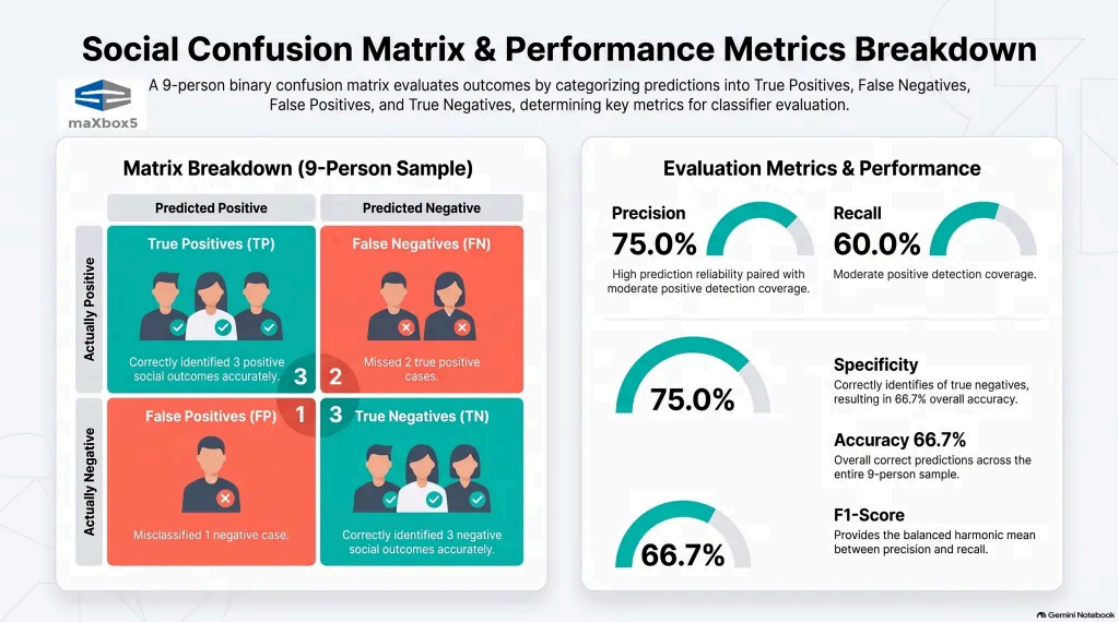
- Sample Prevalence: $\text{Prevalence} = \frac{\text{Actual Positives}}{\text{Total Sample}} = \frac{3 + 2}{9} = \frac{5}{9} \approx 0.5556 \text{ (55.6\%)}$
- Recall: 0.600
- FPR: 0.250

Applying these parameters to the Bayesian transformation formula:

$$\text{Precision} = \frac{0.600 \cdot \left(\frac{5}{9}\right)}{0.600 \cdot \left(\frac{5}{9}\right) + 0.250 \cdot \left(\frac{4}{9}\right)} = \frac{\frac{3.0}$$

{9}}{\frac{3.0}{9} + \frac{1.0}{9}} = \frac{3}{3 + 1} = \frac{3}{4} = 75.0\%

This mathematical equivalence demonstrates that empirical confusion matrix precision represents the Bayesian posterior probability of a case being genuinely positive after observing a positive prediction under a specific sample prevalence.



Conclusion of Confusion Matrix

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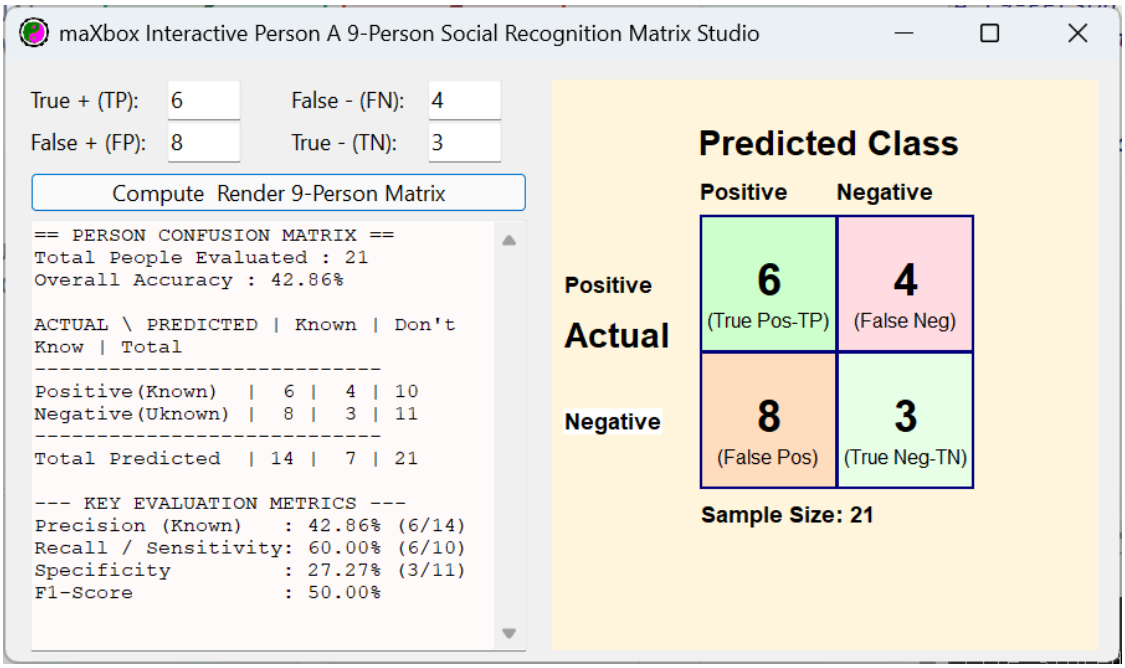
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6. Tools and Links

Scripts on maXbox:

https://sourceforge.net/projects/maxbox5/files/examples/1503_person_the_confusion_matrix2_1interactiv.pas/download

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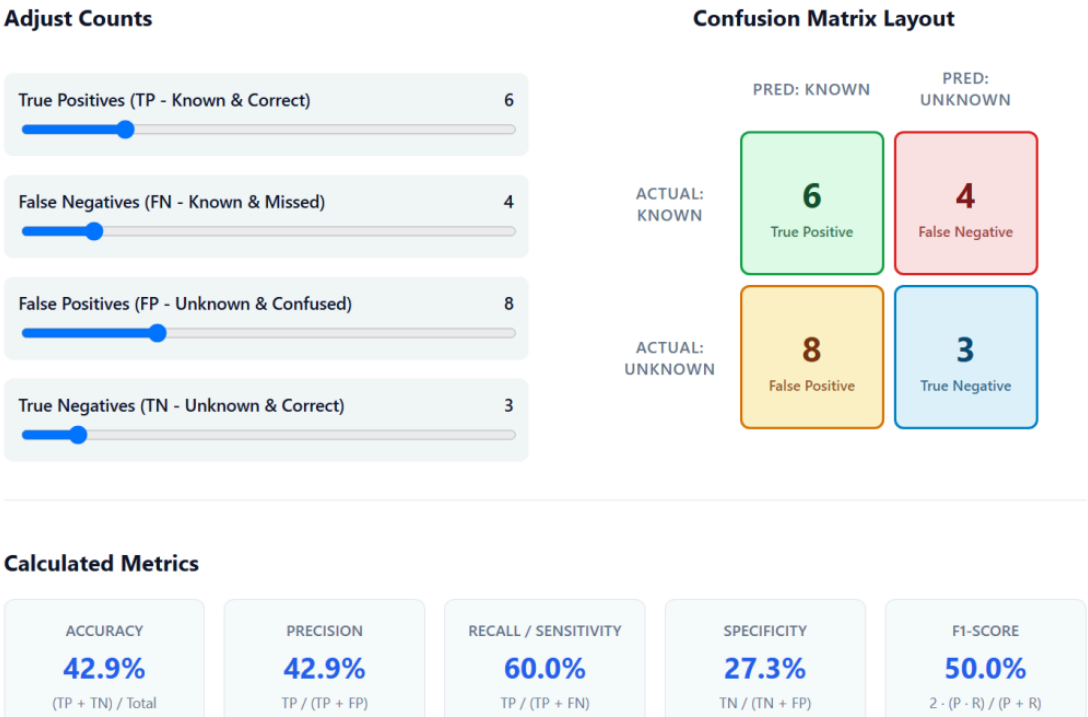
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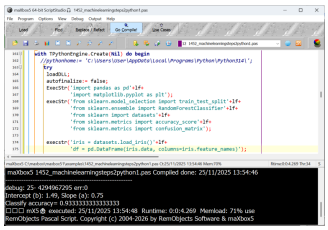
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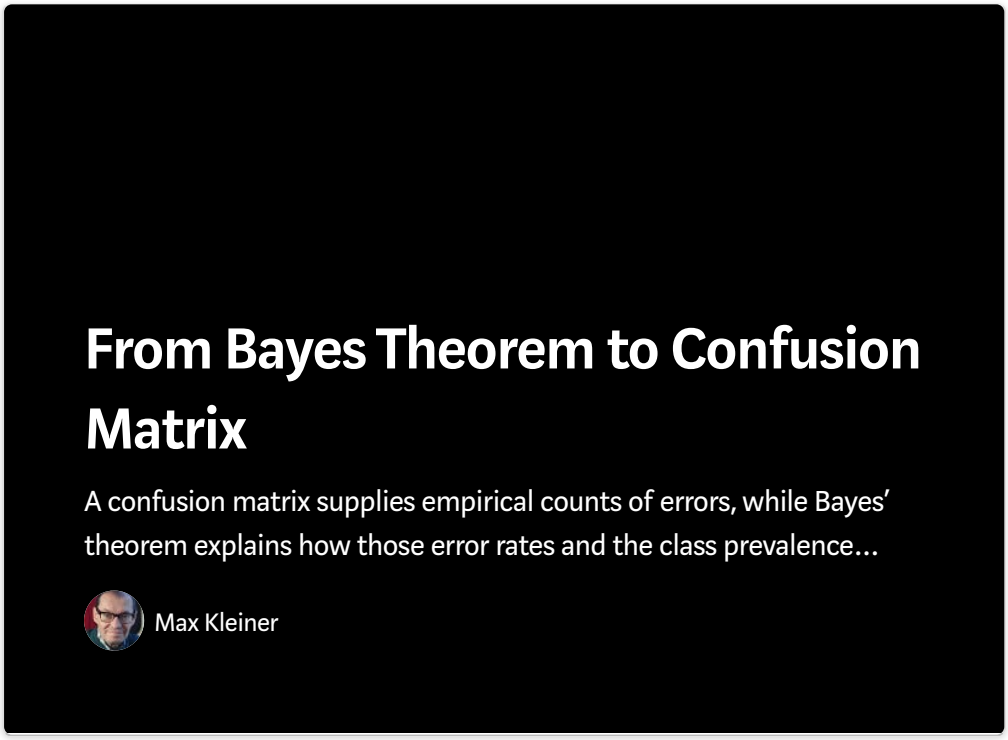
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Machinelearning Steps



This article will provide the clear cut understanding of Iris dataset and how to do classification (supervised learning) on Iris flowers dataset using python, seaborn and sklearn. The Iris dataset is often used in machine learning and data science courses, because it's simple to understand and well-defined, yet interesting enough to present real challenges to … Continue reading

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